

VIVEK RAI

Computer Science Student

Scientific Computing graduate student specializing in **multimodal perception** and behavior modeling. Engineered intersection logic for the **EcoCar CAV** team using RTMaps, Simulink, and RL. Proficient in **Python** pipelines, sensor data analysis, and simulation validation.

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EDUCATION

Master of Science, Computer Science Illinois Institute of Technology

2025 - 2027

Chicago, IL

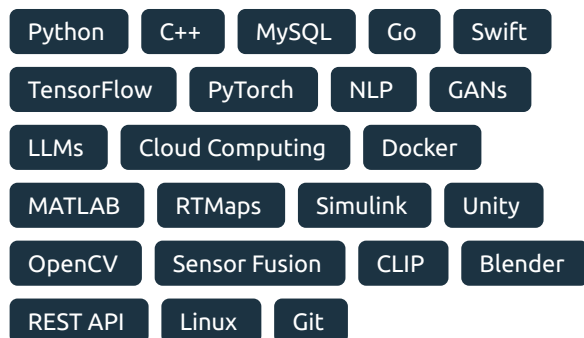
- GPA: 4.0

Bachelor of Technology, Computer Science & Engineering Rajiv Gandhi Institute of Petroleum Technology

2021 - 2025

Raebareli, India

SKILLS / SOFTWARES



CERTIFICATES

Supervised Machine Learning: Regression and Classification (2024)

Stanford University, DeepLearning.AI

Advanced Learning Algorithms (2024)

Stanford University, DeepLearning.AI

Unsupervised Learning, Recommenders, Reinforcement Learning (2024)

Stanford University, DeepLearning.AI

ORGANIZATIONS

Robotics Club (2026 - Present)

Member - NASA Lunabotics Challenge

EcoCar (2025 - Present)

CAV Team Member

DeepLearning.AI (2024 - Present)

Course Tester

IEEE (2022 - 2024)

Editorial Head

Cisco thingQbator (2022 - 2024)

Head of Marketing

WORK EXPERIENCE

CAV Team EcoCAR

09/2025 - Present

Chicago, IL

- Achieved a **95% success rate** with an intersection-handling logic for an Autonomous Vehicle, developed using **Reinforcement Learning (PPO)** and **Simulink**.
- Collaborated with a cross-functional team to implement **sensor fusion pipelines** in **RTMaps**.

VR Intern

Saint Gobain INDEC

05/2024 - 10/2024

Mumbai, India

- Built Unity-based **VR Fire Safety Training App**, saving **20+** staff hours and **\$500+** per VR training session by avoiding physical drills.
- Created and optimized **15+** Blender assets for VR Plant Visit modules.

ML Intern

Saint Gobain INDEC

06/2023 - 08/2023

Mumbai, India

- Created an **NLP-based sentiment analysis system** using Google Cloud Vertex AI, achieving **92%** accuracy.
- Optimized transformer and ML models, reducing inference latency by **40%** and enhancing real-time data analytics, enabling HR and marketing to extract new, accurate insights from text **data at scale**.

PERSONAL PROJECTS

Privacy-Preserving Decentralized Learning for Medical Imaging (08/2025 - 12/2025)

- Engineered a custom **Cross-Silo Decentralized Learning** orchestration pipeline capable of **simulating large hospital networks**.
- Validated **privacy-preserving medical AI** over **100+** experiments by achieving **~80% accuracy** on PathMNIST, retaining **94%** of centralized baseline performance while preventing data leakage.

Football Match Analysis (04/2025 - 05/2025)

- Led a team to create the tool to get **live data analytics** from football match videos.
- Built **real-time** tracking + visualization pipeline at **30 FPS** using **YOLOv5** and **OpenCV**.
- Annotated **2,000+** frames with Roboflow and debugged tracking failures.

Visual Question Answering (10/2023 - 12/2023)

- Built a **multimodal AI system** integrating CLIP embeddings and **LSTM-based** language modeling for natural language question answering over images.
- Implemented **attention mechanisms** and **token embeddings** to improve alignment between textual and visual features, enabling visually impaired users to receive answers with **85% accuracy**.